Question 1: TE/TM modes in PEC waveguides (30 marks)

Consider a uniform electromagnetic waveguide extending along the z-direction. Develop a numerical simulation tool to determine the propagating modes of a waveguide with a two-dimensional cross-section bounded by perfect electric conductor (PEC) walls.

Solver requirements (TE and TM modes)

1. Define the waveguide cross-section using a Gmsh mesh. The geometry should not be restricted to a rectangular cross-section.
2. Implement PEC boundary conditions on user-specified boundaries. PEC boundaries may be located at arbitrary boundaries of the waveguide cross-section.
3. Determine the supported TE and TM modes and calculate their propagation constants β and cutoff frequencies.
4. Obtain the corresponding electric and magnetic field distributions and, for selected modes, plot the longitudinal field components and the transverse electric and magnetic field vectors.
5. Validate the numerical results using a rectangular PEC waveguide by comparing the calculated propagation constants and cutoff frequencies with analytical results.

Analysis and reporting

The student must:

1. clearly state the physical assumptions and field representation used;
2. derive the governing equations for the TE and TM modes starting from Maxwell's equations, and derive the boundary conditions imposed by the PEC walls;
3. describe the numerical formulation used to solve the resulting problem and justify the choice of numerical method; and
4. compare the numerical results with analytical results for the rectangular PEC waveguide.

Deliverables

1. Source code of the numerical solver. It should take a mesh file as input and output the field plots, propagation constants, and cutoff frequencies of selected modes.
2. Derivation of the TE/TM waveguide formulation.
3. Field plots for selected TE and TM modes in the report.
4. Rectangular-waveguide validation results and error analysis in the report.
5. A brief discussion of the obtained results in the report.

Evaluation note.

- For the main question, the choice of numerical method is left to the student. Any appropriate numerical approach may be used, provided that the formulation is properly derived and the numerical results are physically justified.
- The submitted code will be tested on arbitrary waveguide shapes. These test cases constitute **30% of the marks for this question** (not bonus). For each test case, the code must produce

- the propagation constants and cutoff frequencies of selected modes,
- a transverse field plot showing the electric and magnetic field vectors, and
- a heatmap of the real parts of the longitudinal field components at $z = 0$.

test cases.

Bonus question (5 marks)

Extend the solver developed in the main question to analyse a more general waveguide problem (uniform along z) supporting multiple dielectric regions, with each region having independently specified ϵ_r and μ_r. The extended solver must not assume that either E_z or H_z is zero; both longitudinal components may be non-zero.

The solver should be capable of handling arbitrary waveguide geometries and material distributions and should determine the corresponding electromagnetic modes and field distributions.

For the bonus question, the choice of numerical method is also left to the student, provided that it is capable of handling the general full-vector problem.

Question 2: L-type matching network design using the Smith chart (30 marks)

A lossless L-type matching network can be used to transform a complex load impedance to a specified source impedance. An L-network consists of two reactive elements, one connected in series and the other in shunt. The Smith chart provides a graphical method for determining the required reactance and susceptance.

The matching network should transform the load to $Z_{\text{in}} = Z_0$, corresponding to the centre of the Smith chart.

Design the L-type matching network for

1. $Z_L = (150 + j45)\ \Omega$;
2. $Z_L = (30 - j15)\ \Omega$;
3. $Z_L = (60 + j75)\ \Omega$.

For all parts, use $Z_0 = 75\ \Omega$ and $f_0 = 1$ GHz.

Frequency response

For each part, investigate the frequency response of the matching network. Assume the source has $R_s = 75\ \Omega$. Calculate and plot, over an appropriate frequency range around f_0,

$$\Gamma_{\text{in}}(f) \qquad \text{and} \qquad 20\log_{10}\left|\frac{V_{\text{out}}(f)}{V_s(f)}\right|.$$

Identify the frequency at which maximum matching occurs and determine the corresponding -3 dB bandwidth of the transfer response.

Bonus (5 marks)

Repeat one of the matching problems using only the impedance Smith chart. You may not directly use the admittance half of a ZY Smith chart. Determine how the required shunt reactive element can be designed using only impedance transformations, and compare the procedure with the impedance–admittance Smith chart solution.

Question 3: Transient response of a switched transmission line (30 marks)

Consider the circuit in Fig. 1. A 1 V DC source ($R_s = 25\ \Omega$) drives a 20 m transmission line terminated in $R_L = 100\ \Omega$. Switch S closes at $t = 0$ and opens at $t_1 = 250$ ns, open-circuiting the source end. Line parameters are provided in Table 1.

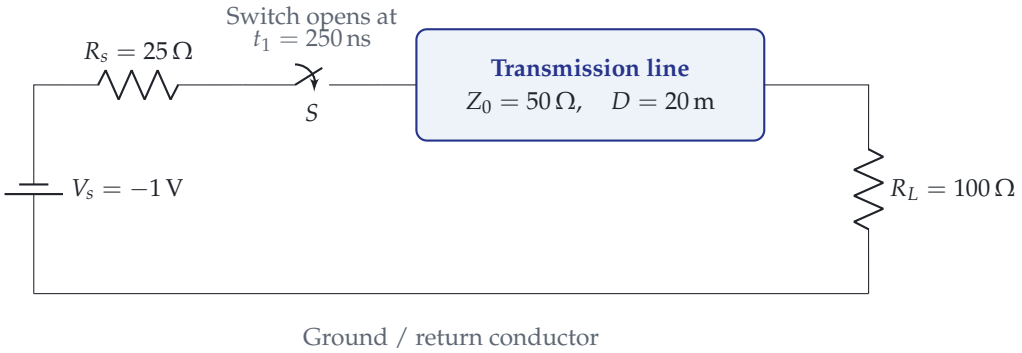


Figure 1: Switched transmission-line circuit.

Table 1: Transmission-line parameters.

Parameter	Value	Description
L'	250 nH/m	Inductance
C'	100 pF/m	Capacitance
R'	0.15 Ω/m	Series resistance
G'	10^{-7} S/m	Shunt conductance
D	20 m	Line length

Assignment tasks

1. Calculate Z_0 and the propagation velocity v_p. Compare v_p with the speed of light to evaluate physical validity.
2. Determine Γ at the load, and at the source for both switch states (closed and open). State the corresponding boundary conditions.
3. **Bounce diagram for the lossless line.** Construct a complete bounce diagram accounting for the switch opening at $t = 250$ ns. Label the initial wave, subsequent reflections, amplitudes, and arrival times. Plot $V(t)$ at the source and the load.
4. **FDTD simulation.** Build a numerical FDTD model explicitly modelling the switch states for three scenarios:
   (a) lossless nondispersive ($R' = G' = 0$);
   (b) lossy nondispersive (using Table 1);
   (c) dispersive, modelling the dielectric using the Debye relation

$$\epsilon_r(\omega) = \epsilon_\infty + \frac{\epsilon_s - \epsilon_\infty}{1 + j\omega\tau}, \qquad C'(\omega) = C'_0 \frac{\epsilon_r(\omega)}{\epsilon_{r,0}},$$

   where $\epsilon_s = 2.56$, $\epsilon_\infty = 2.25$, and $\tau = 5$ ns.

Required simulation deliverables.

- Runnable source code that reproduces all outputs.
- Real-time animation of the voltage wave along the line.
- Time-domain plots comparing $V(t)$ at the source, the load, and an interior point.
- Analysis of pulse broadening, shape distortion, and frequency arrival times due to dispersion.

5. **FDTD versus analytical.** Compare the numerical FDTD results against the theoretical lossless bounce-diagram predictions. Discuss agreement in timing and amplitudes, and identify any numerical artefacts arising from discretization.